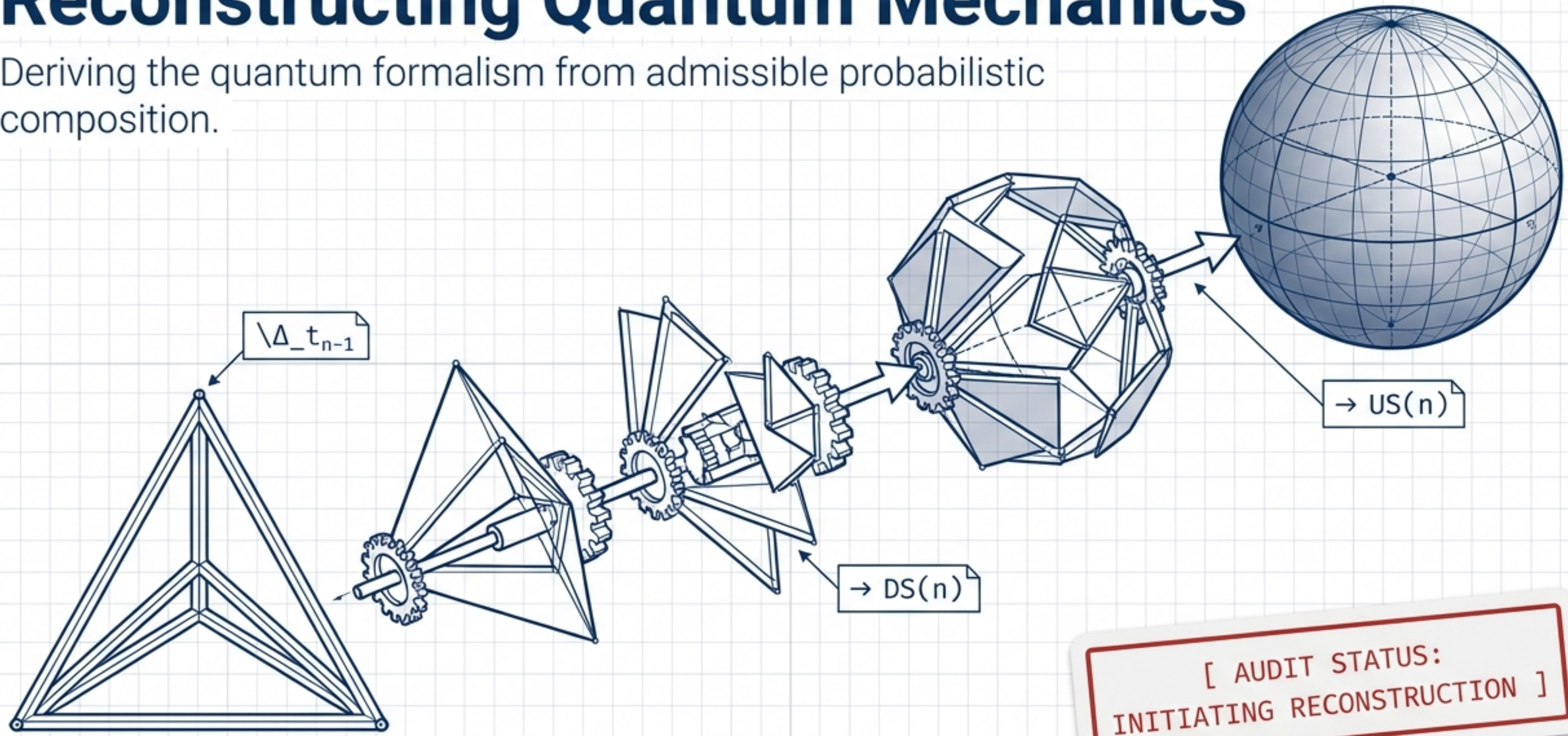


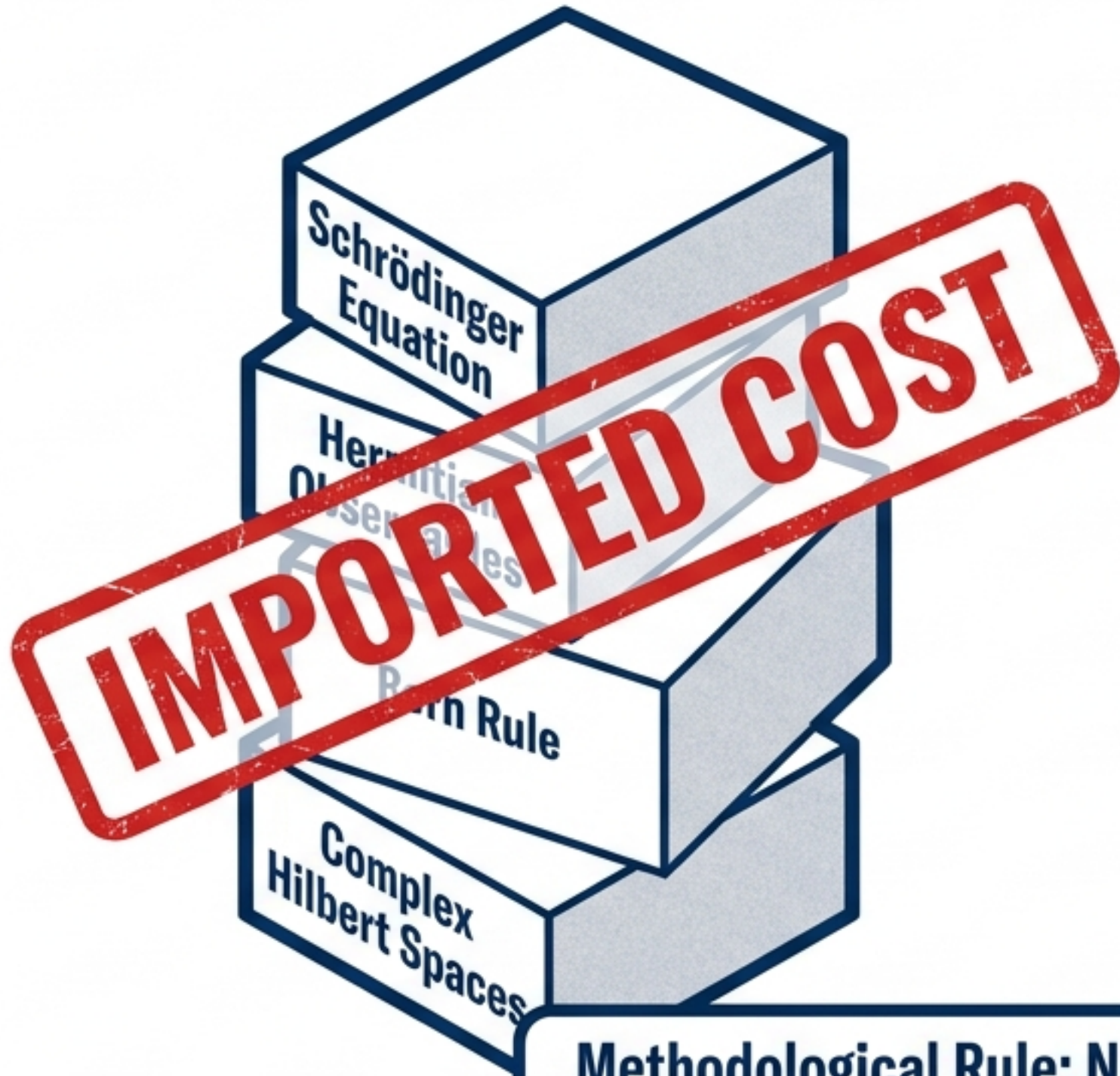
Reconstructing Quantum Mechanics

Deriving the quantum formalism from admissible probabilistic composition.

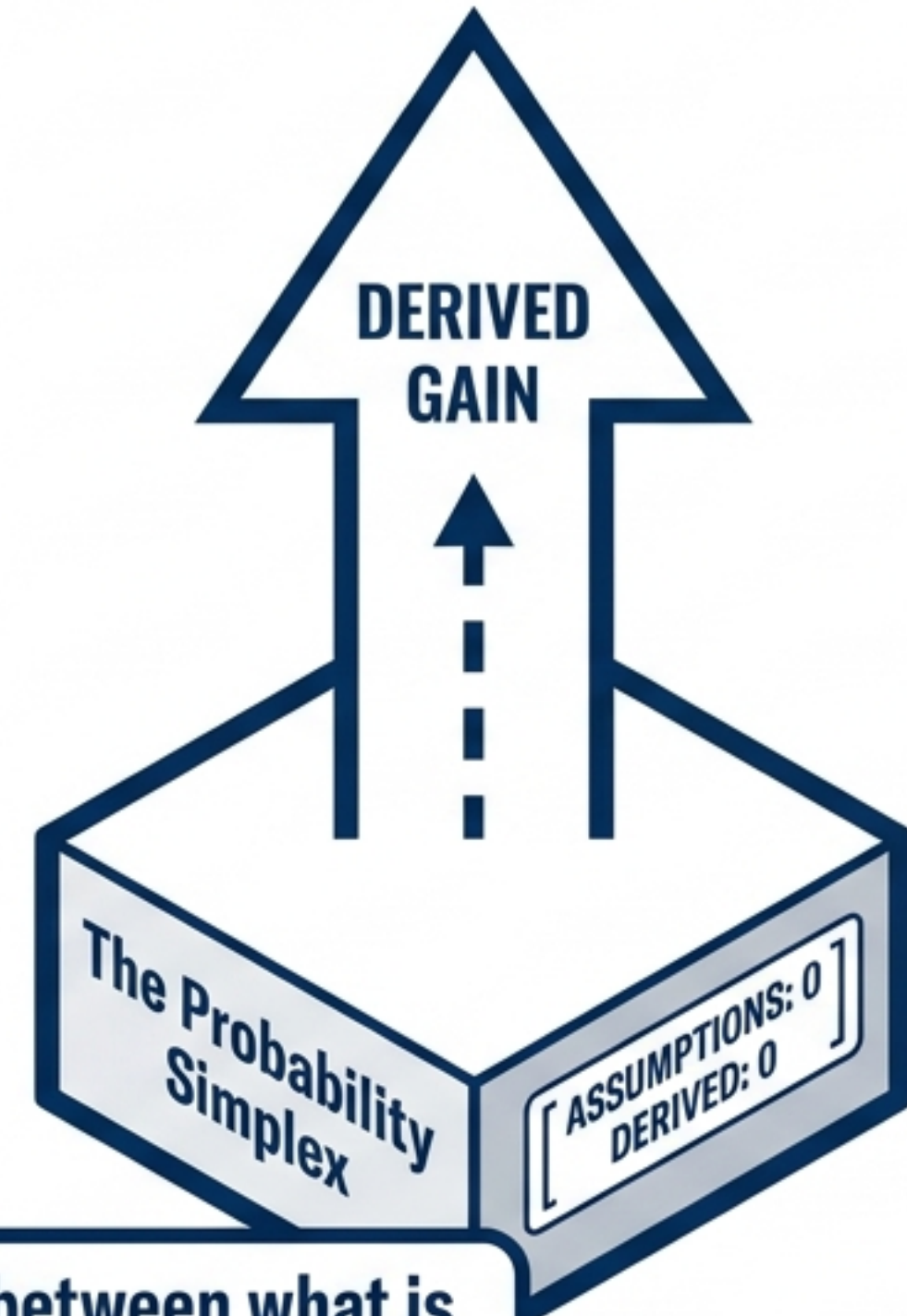


Standard quantum mechanics relies on heavy postulates. We reconstruct it from bare probability.

Standard QM

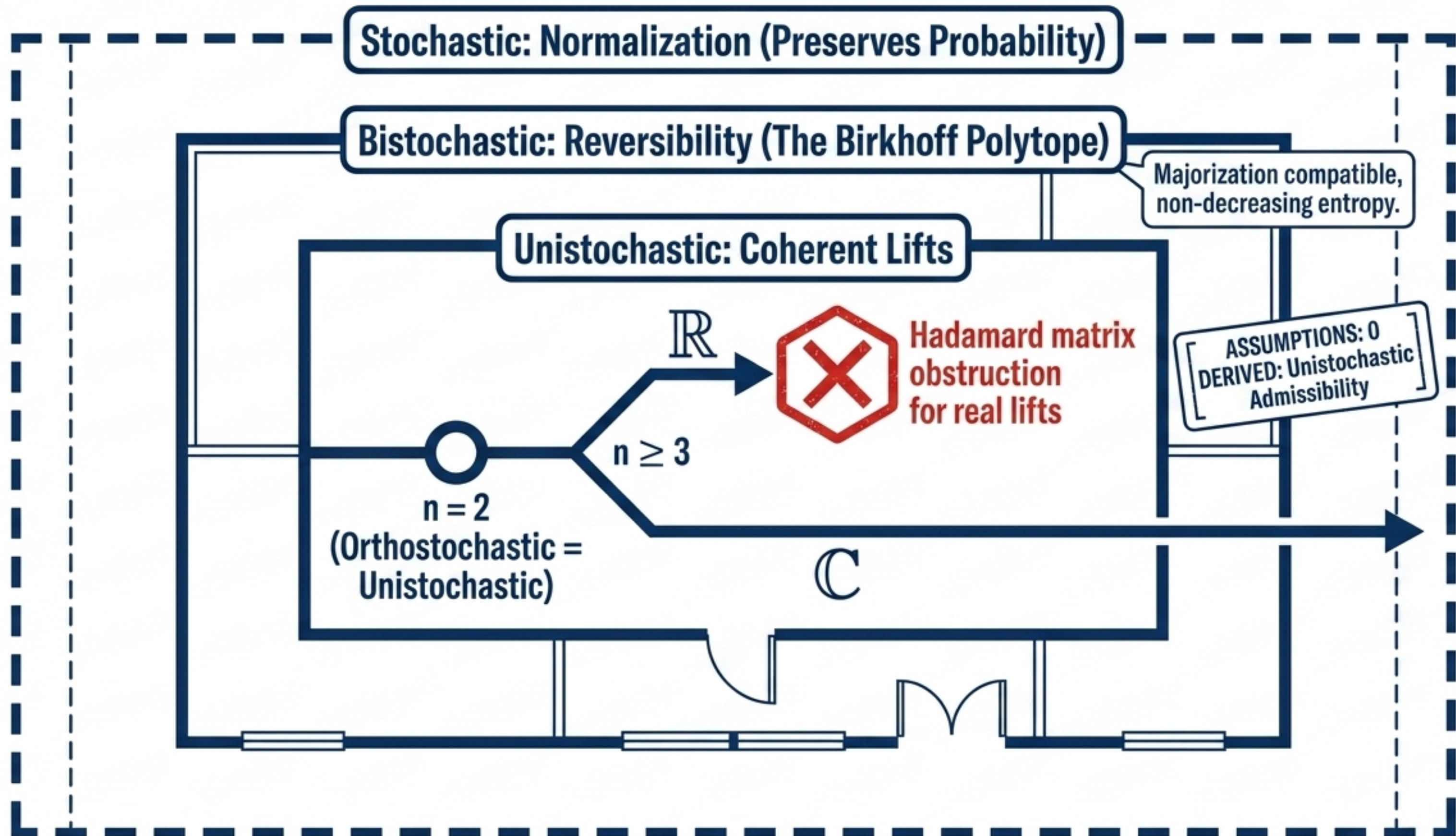


The Reconstruction



Methodological Rule: Never blur the lines between what is derived from constraint and what is assumed by postulate.

Reversibility restricts probabilistic maps, splitting real and complex theories at $n \geq 3$.



Probability-level composition discards structural information retained by amplitudes.

The Interference Gap

Path A: Amplitude-Level Composition



$$|(U_2U_1)_{ij}|^2 = \sum_k |(U_2)_{ik}|^2 |(U_1)_{kj}|^2 + \sum_{k \neq l} (U_2)_{ik} (U_1)_{kj} (U_2)_{il}^* (U_1)_{lj}^*$$

Path B: Probability-Level Composition



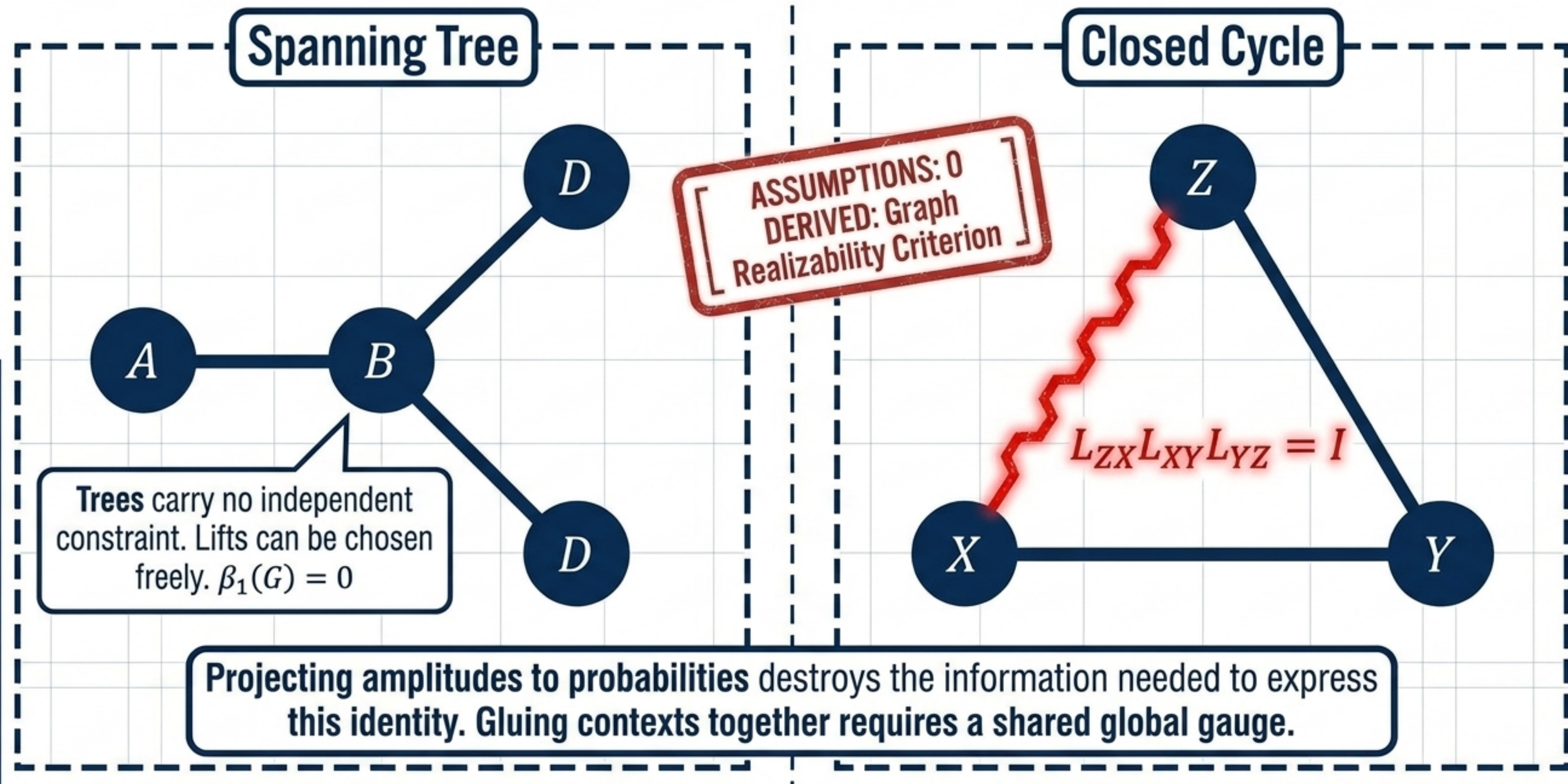
$$|(U_2U_1)_{ij}|^2 = \sum_k |(U_2)_{ik}|^2 |(U_1)_{kj}|^2 + \sum_{k \neq l} (U_2)_{ik} (U_1)_{kj} (U_2)_{il}^* (U_1)_{lj}^*$$

A coherent lift restores compositional closure.
Note: Interference alone does not prove complex numbers; real lifts exhibit the same gap.

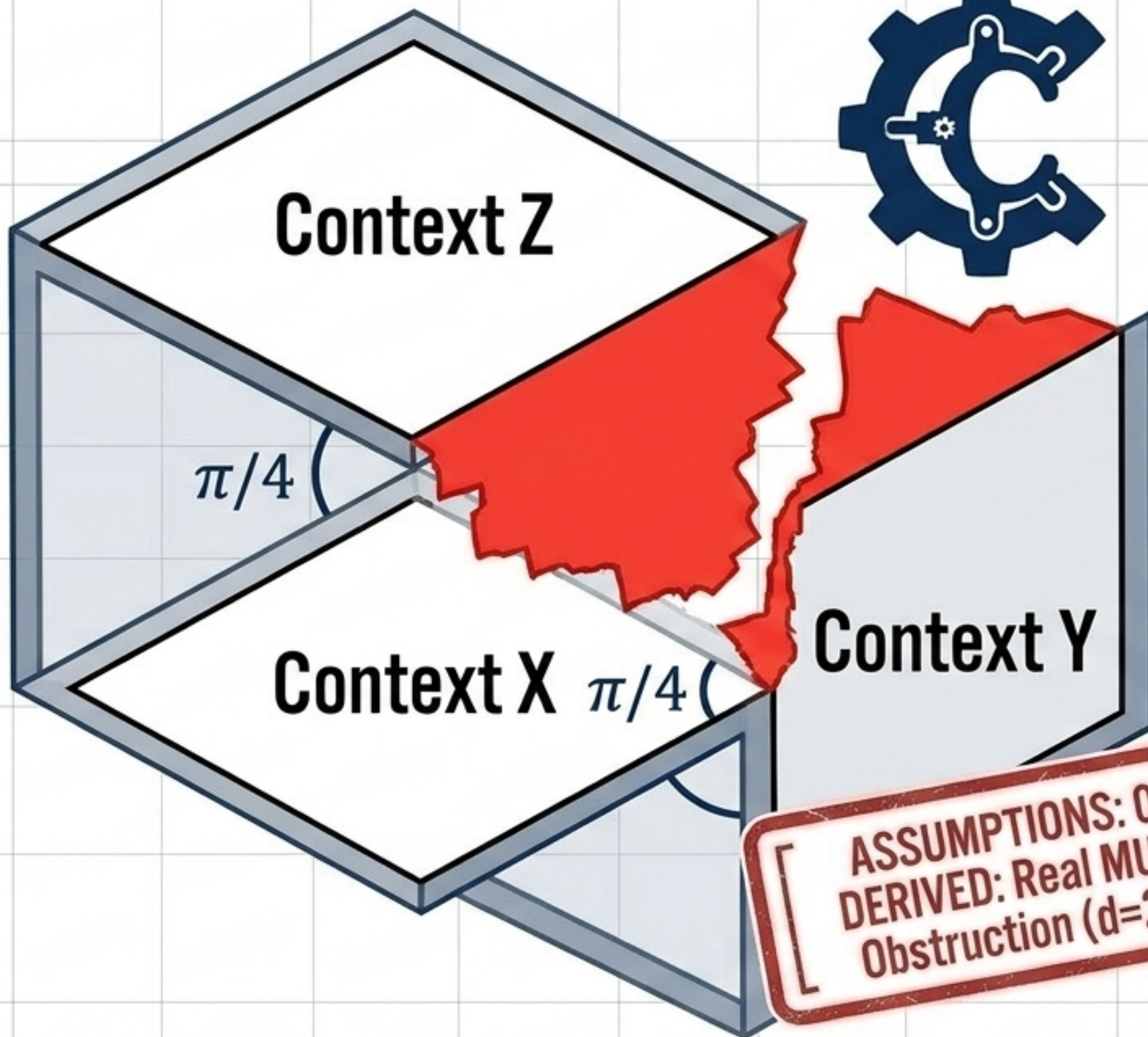
Lost
Interference
Data

ASSUMPTIONS: 0
DERIVED: Compositional
Obstruction

Closed loops of transitions force global gauge consistency checks.



Real numbers structurally fail to glue three mutually unbiased bases.



Triangle-Gluing Theorem:

$$\theta_{XY} \equiv \pm\theta_{ZX} \pm \theta_{ZY} \pmod{\pi/2}$$

For $\mathbb{R}^2$ MUBs, this requires:

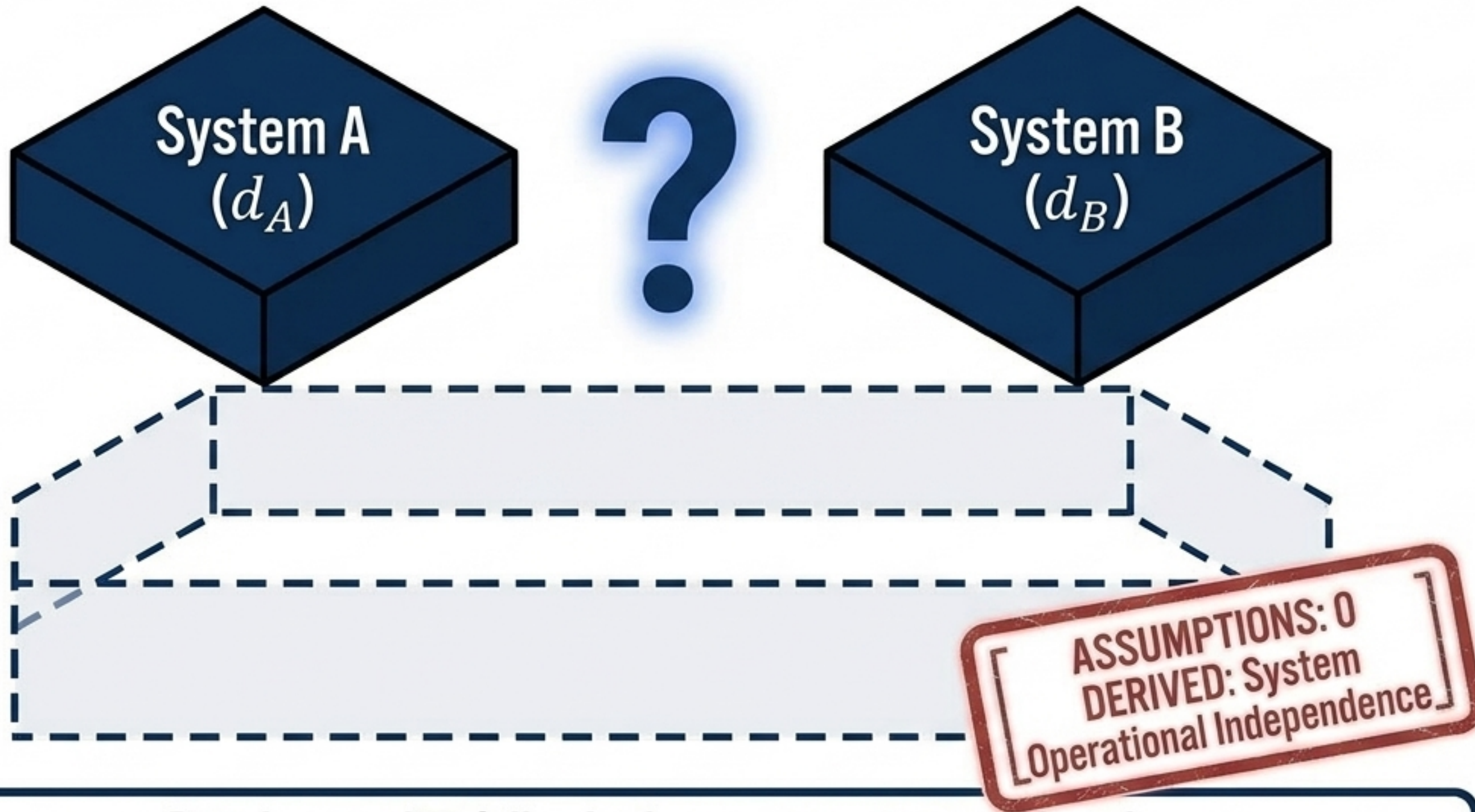
$$\pi/4 \equiv 0 \pmod{\pi/2}$$

A mathematical contradiction.

**Realizability over $\mathbb{R}$
is measure-zero once
cycle rank ≥ 1 .**

ASSUMPTIONS: 0
DERIVED: Real MUB
Obstruction (d=2)

How must independently accessible systems compose?



Part I proved $\mathbb{R}$ fails single-system contextual cycles.
But what happens when we construct joint descriptions of independent systems?
We require a theory where the composite does not obscure the parts.



$\mathbb{R}$ (Real)



$\mathbb{C}$ (Complex)



$\mathbb{H}$ (Quaternionic)

Local Tomography strictly isolates Complex numbers as the only viable scalar field.

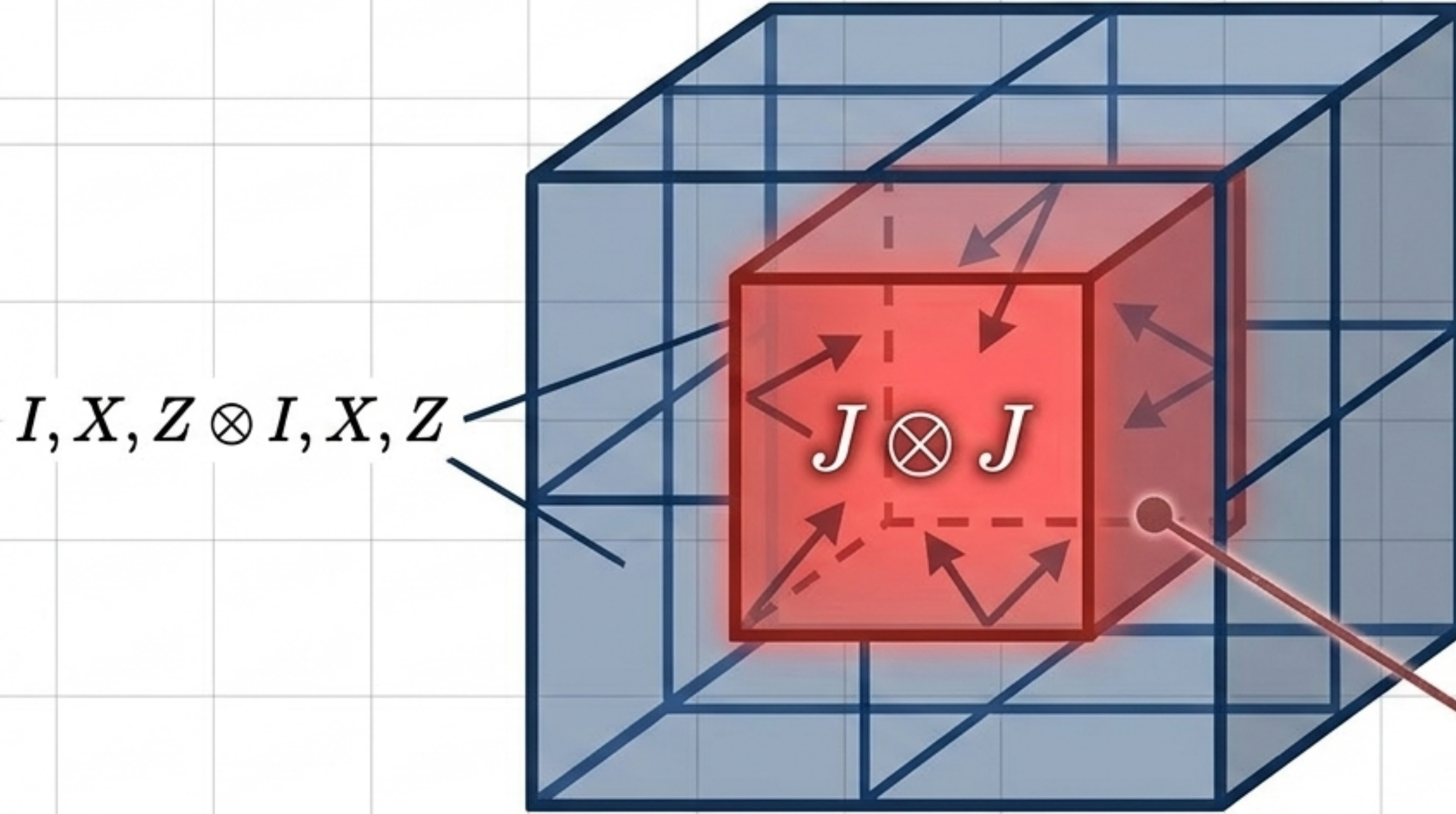
[[A] ASSUMPTION 1:
LOCAL TOMOGRAPHY]

Field	Local Product $\beta_F(d_A) \times \beta_F(d_B)$	Actual Joint Space $\beta_F(d_A d_B)$	Audit Status
$\mathbb{R}$ (Real)	9 (for $d_A=d_B=2$)	10	FAILS (Missing Parameters)
$\mathbb{H}$ (Quaternionic)	36	28	FAILS (Overdetermined)
$\mathbb{C}$ (Complex)	16	16	MATCHES EXACTLY

[COST: 1 Postulate
EARNED: The Complex
Tensor Product]

Real composites harbor a strictly nonlocal, locally invisible degree of freedom.

Real 2-Qubit System



$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

J is antisymmetric
(locally unobservable over $\mathbb{R}$).

But $J \otimes J$ is symmetric
(a valid global observable).

This exact dimension is annihilated by the partial trace:
 $\text{tr}_A(J \otimes J) = 0$. Globally physical, locally unrecoverable.

COST: 1 Postulate
EARNED: The Missing
Real Dimension

Separability is not intrinsic to a state; it is field-relative.

$$\rho(c) = \frac{1}{4}(I \otimes I + cJ \otimes J)$$

Entangled for all $c \neq 0$.

Evaluated in $\mathbb{R}$

$c = -1$

$c = 0$

$c = 1$

[[COST: 1 Postulate] | [EARNED: Field-Relative Entanglement]

$$\rho(c) = \frac{1}{4}(I \otimes I + cJ \otimes J)$$

Evaluated in $\mathbb{C}$

$c = -1$

$c = 0$

$c = 1$

Completely separable across its entire physical range.

The complex theory reveals the identical physical matrix as classically correlated, not entangled.

Decomposes into four pure product states via
 $J \otimes J \rightarrow -\sigma_y \otimes \sigma_y$

The Born Rule is a derived hierarchy, not a singular postulate.

Layer 1: Context-Transition

$$B_{ij} = |U_{ij}|^2$$

(Derived from unistochastic logic)



[A] ASSUMPTION 2: Pure states are frame-completable vectors.

Layer 2: State Vector Rule

$$p_i = |\langle e_i | \psi \rangle|^2$$

(Derived)

COST: 2 Postulates
EARNED: The Value:
The Born Rule

Identifying a 'state' as a distinguished coordinate of a measurement frame recovers the textbook Born rule at zero additional cost.

Layer 3: Density Operator Rule

$$p_i = \text{tr}(\rho P_i)$$

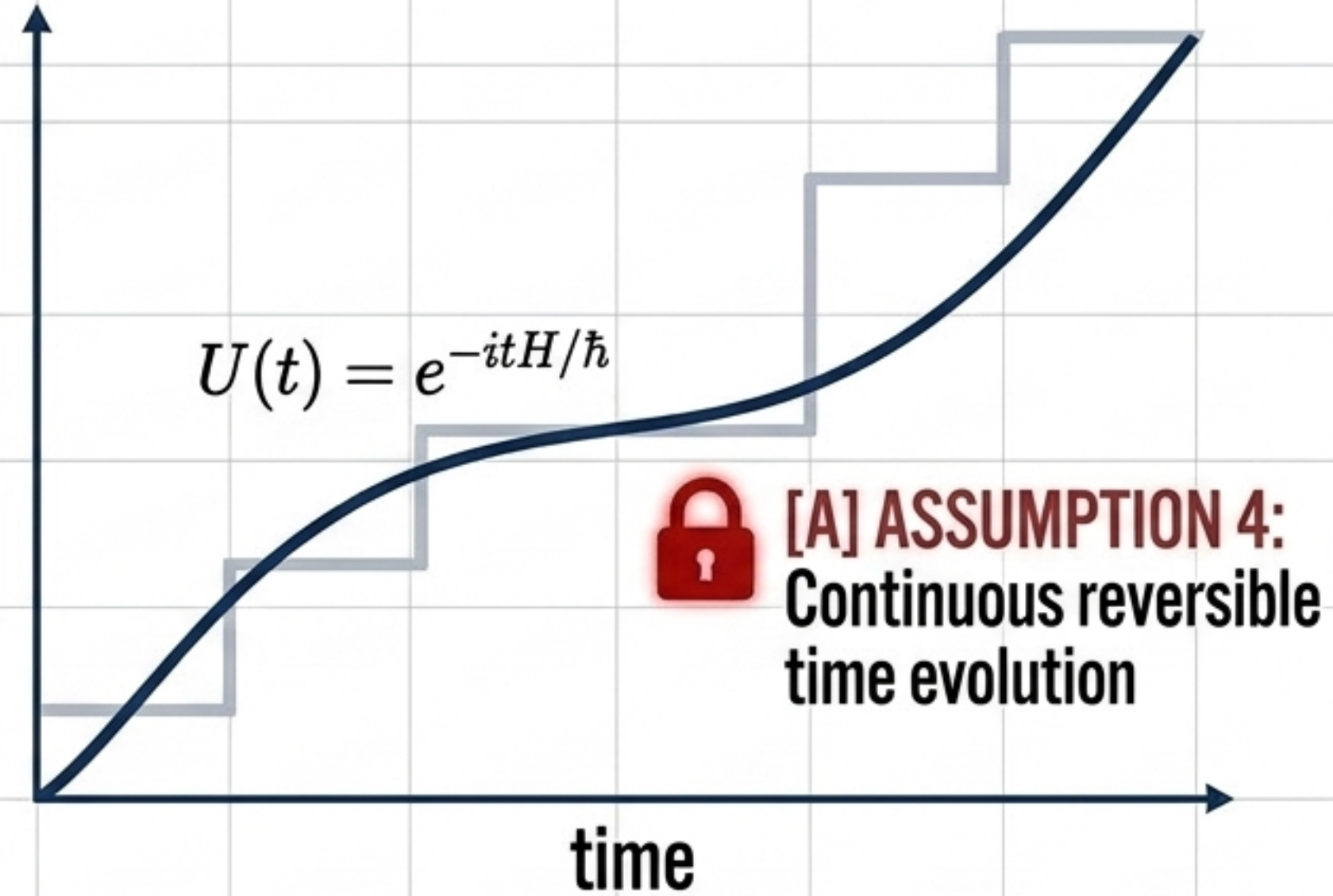
(Derived via affine mixing)

Observables and continuous dynamics emerge from operational packaging.

Observables



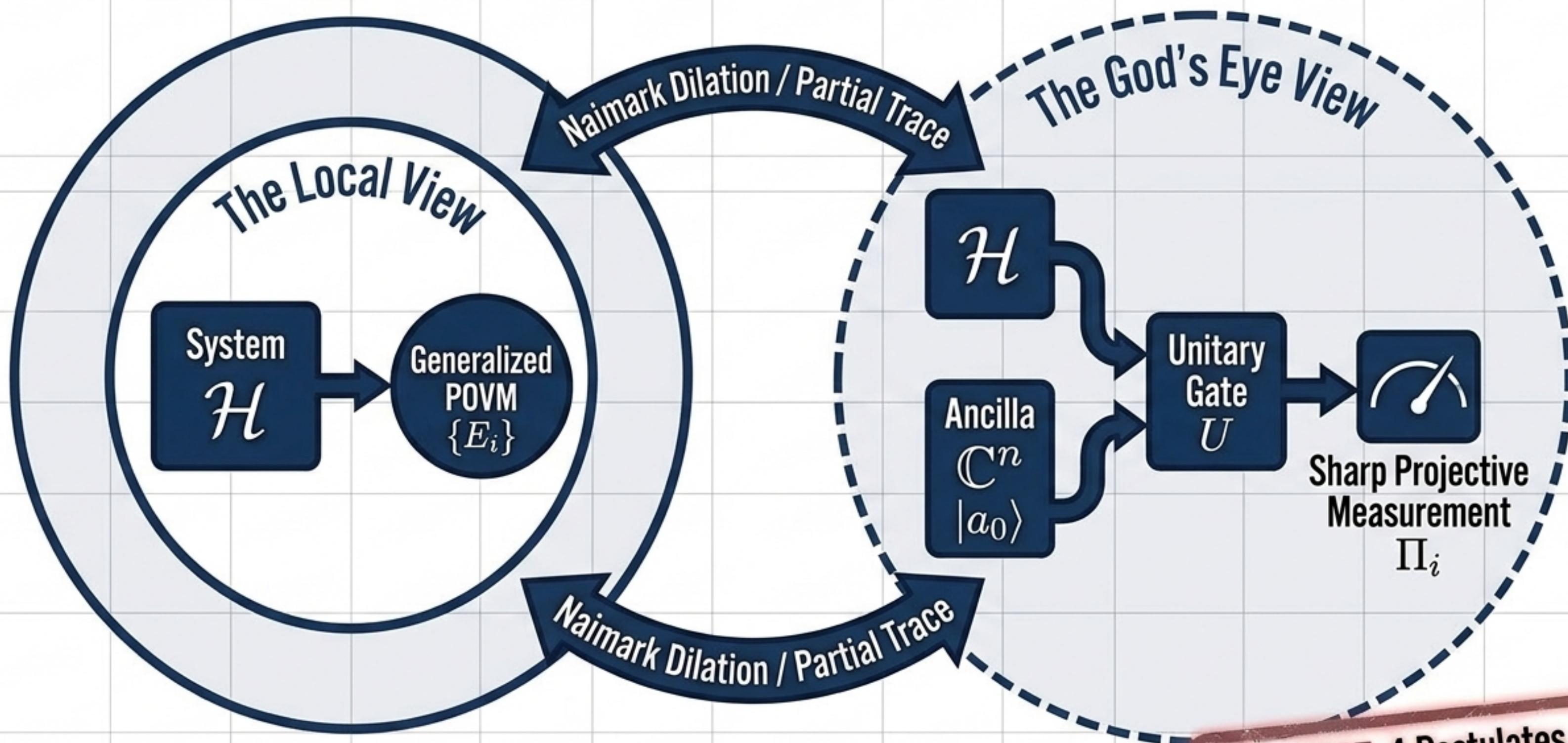
Dynamics



Schrödinger's equation ($i\hbar\partial_t|\psi\rangle = H|\psi\rangle$) is simply the infinitesimal generator of continuous composition. It is not an independent postulate.

COST: 4 Postulates
[EARNED: Hermitian Observables & Schrödinger]

Generalized measurement (POVM) is simply a sharp measurement, elsewhere.

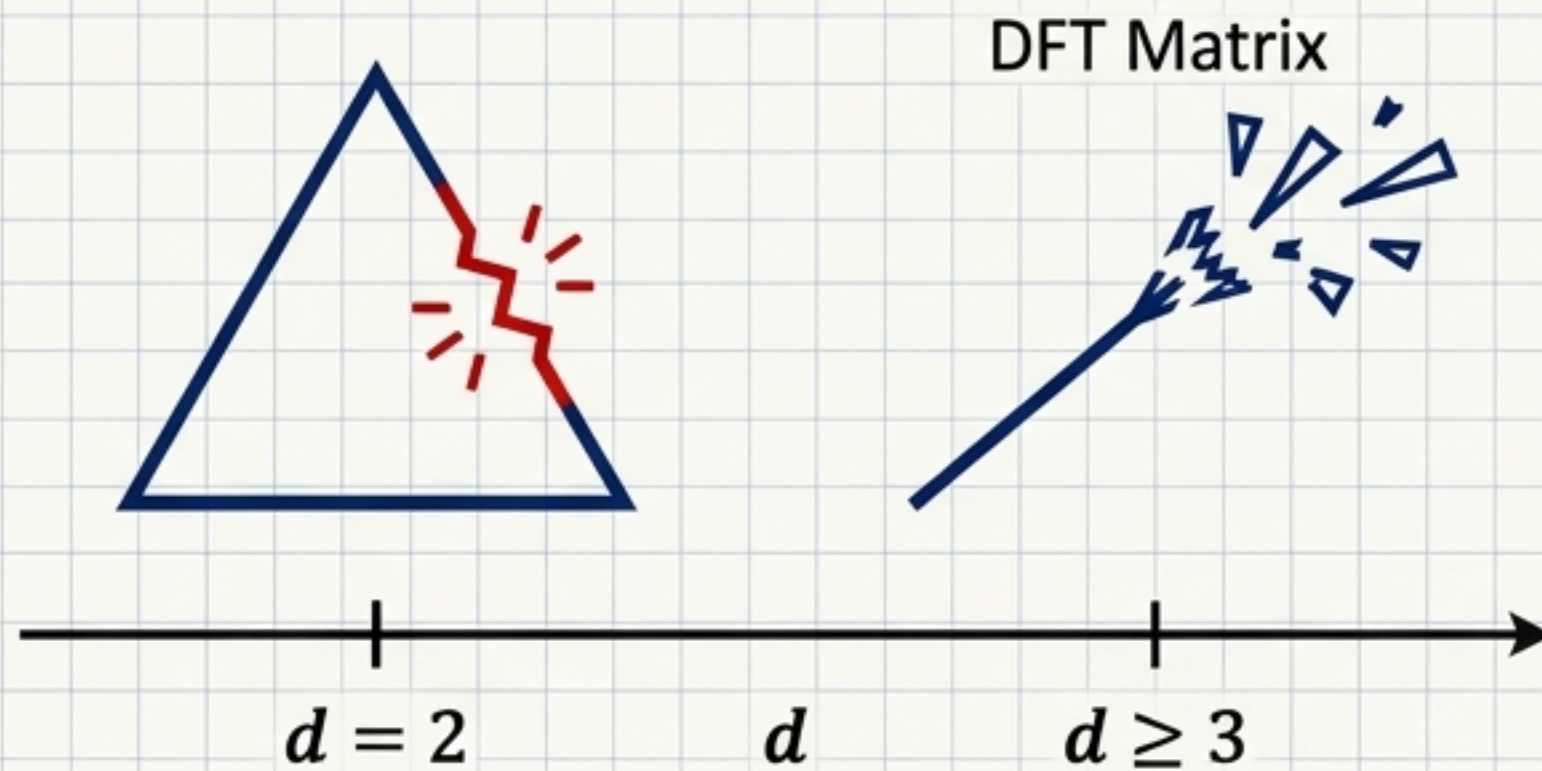


The POVM formalism requires no new physics. It relies on the identical system-plus-ancilla architecture introduced to manage entanglement.

COST: 4 Postulates
[EARNED: Generalized Measurements (POVMs)**]**

Structural Deepening: Higher dimensions and the Amplitwist.

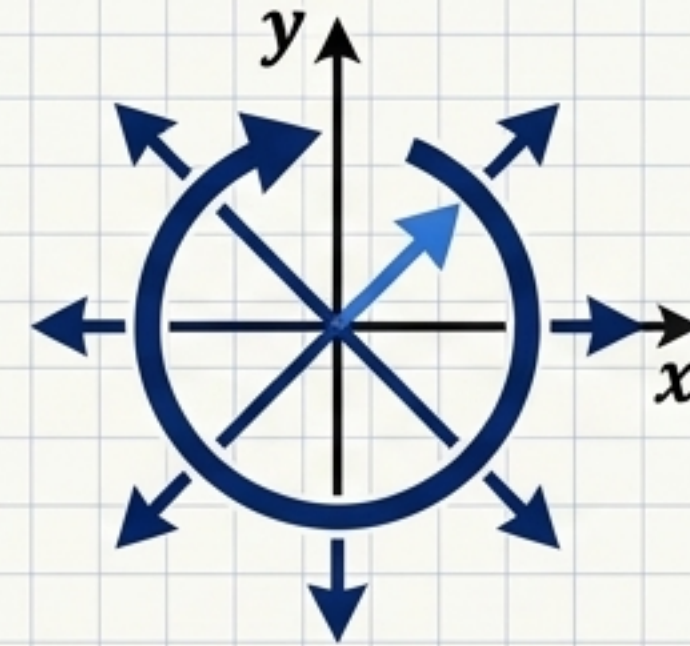
The Dimension Collapse



At $d \geq 3$, single edges already witness real-inadmissibility. The minimal witness collapses.

The Amplitwist Identity

$$i = J$$



Complex arithmetic is just **amplitwist** composition of real 2×2 matrices. The actual obstruction to a purely Real quantum theory is the **composition rule** for **phase-redistribution**, not the scalar field itself.

[**COST:** 4 Postulates] | [**EARNED:** Higher-D Structural Clarity]

The Final Audit: A massive architectural yield for a minimal cost.

Balance Sheet

**FINAL AUDIT
COMPLETE**

The Assumed Cost

1. Local Tomography
2. Pure states are frame-completable vectors
3. Measurement outcomes are labeled by real numbers
4. Time evolution is a continuous unitary group

The Derived Structure

- ✓ Complex Necessity ($\mathbb{C}$)
- ✓ Born Rule Hierarchy
- ✓ Hermitian Observables & Spectral Decomposition
- ✓ Schrödinger's Equation
- ✓ Entanglement & Partial Trace
- ✓ POVMs via Naimark Dilation

Quantum mechanics is not an arbitrary set of historical postulates; it is the inevitable mathematical consequence of structural coherence in probabilistic composition.